

Phoneix Project CyberSecurity Department (CY003)

Introduction

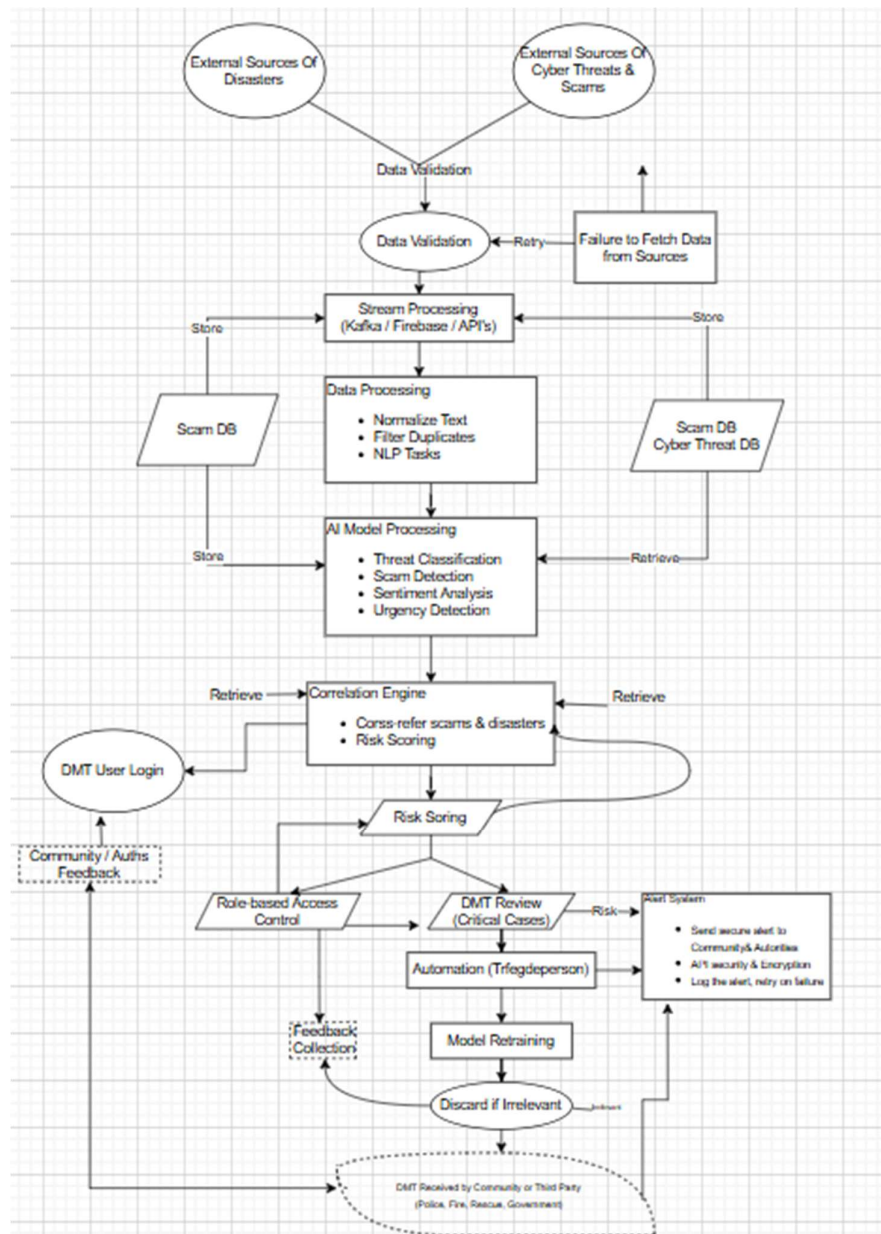
Natural disasters in Australia no longer pose purely physical risks. As communities increasingly depend on digital infrastructure for emergency communication, financial assistance and information sharing. Disaster events have become one of the most exploited opportunities in the cybercriminal playbook. When a disaster strikes, public attention is used by survival and that disturbance creates a window that malicious actors have repeatedly and severely taken advantage of. The evidence from Australian disaster events makes this reality impossible to ignore and the scale and variety of cyber threats that emerge during these periods continues to grow with each passing year. (Scamwatch, 2020; Chirgwin, 2023; Elsworthy, 2022).

Project Phoenix was built on the recognition that protecting communities during a disaster means protecting them on two fronts simultaneously, the physical and the digital. The cybersecurity objectives of this framework are therefore centred on one core ambition: to ensure that no disaster event in Victoria becomes an open season for those who seek to exploit it. This means detecting threats the moment they emerge, understanding them in the context of the disaster unfolding in real time, responding to them with both speed and human accountability, and continuously strengthening the system's capabilities so that it stays ahead of an adversary landscape that never stands still.

To achieve this, the Phoenix cybersecurity framework is created around five interconnected goals. The first is real-time threat detection, identifying cyber threats and malicious activity as they emerge during active disaster events, rather than hours or days later when the damage has already been done. The second is contextual intelligence, correlating detected threats against live disaster event data so that attack patterns specifically designed to exploit a disaster are recognised for what they are. The third is balanced response, combining the speed of AI-driven automation with the judgement of trained human analysts, so that high-stakes decisions always carry accountability. The fourth is continuous improvement building a system that learns from every disaster event and every detected threat, growing stronger over time rather than becoming obsolete. The fifth is trusted communication, ensuring that every official alert Phoenix helps to deliver is protected against interception, spoofing, and suppression, so that communities can trust what they receive when it matters most.

These five goals shape every architectural decision described in the sections that follow.

DFD Flow Chart



Let' understand the DFD Chart First

A. External Sources of Disasters

Phoenix started by ingesting real-time disaster intelligence from authoritative monitoring agencies. In Victoria, primary sources consists of the Bureau of Meteorology (BoM) for weather and flood alerts, the Country Fire Authority (CFA) and

Emergency Management Victoria for bushfire updates, and Geoscience Australia for seismic data, all delivered via REST APIs.

Within the cybersecurity architecture, this component does more than provide situational awareness. It acts as a dynamic contextual trigger. The moment, the disaster is detected in a region, Phoenix automatically upgrades its cyber threat monitoring sensitivity for that area. The types of cyber threats that show up during a disaster are not random, they are shaped by the nature of the event itself and they appear with a speed that only a system fixed to live disaster intelligence can match (Chakraborty et al., 2024).

B. External Sources of Cyber Threats & Scams

Running in parallel with the disaster data stream is a continuous feed of real-time cyber threat and scam intelligence. Key sources consists of the ACCC Scamwatch database, the Australian Signals Directorate (ASD) threat feeds, the MITRE ATT&CK framework for known adversary tactics and techniques and social media platforms containing Twitter/X and Facebook monitored for misinformation and fraudulent activity.

What makes this component powerful is not what it provides in isolation but what it enables in combination with the disaster data. A suspicious post or scam report viewed alone may appear unremarkable. Viewed against the backdrop of an active disaster in the same region, as happens in the Correlation Engine, that same content becomes a highconfidence threat signal demanding immediate response. This integration of live cyber threat intelligence with real-time disaster context is the foundational innovation of Phoenix (Husari et al., 2017; Ghazi et al., 2018).

C. Data Validation

The data passes through Data Validation when no data is entered in the Phoenix pipeline. It's the system's first line of defence which serve two critical purposes.

Quality assurance is the first one. Schema validation verified expected fields are present and correctly structured. Format checking check data types match expected values. Deduplication make sure that if the same content is not processed multiple times, a common problem during disasters when the same alert or scam report is submitted through multiple channels simultaneously.

The second point is about security. Phoenix treats every external data feed as untrusted until it is verified. To check if a source is legitimate, it uses API keys and digital certificates. This helps Phoenix tell the difference between real data sources and fake ones. This also protects against a type of attack where someone secretly tampers with a data source before the information even reaches Phoenix. If a data feed gets compromised, it could hide real warnings or trigger false ones. Either way, people would stop trusting the system, which would defeat its entire purpose.

The component also contains an automatic retry mechanism. If due to network disruption, data cannot be fetched then, API downtime or a denial-of-service attack, Phoenix raises a Failure to Fetch flag and retries using exponential backoff, maintaining monitoring continuity even when infrastructure is under stress.

D. Stream Processing (Kafka / Firebase / APIs)

Validated data enters the Stream Processing layer, which routes and delivers continuous data flows to all downstream systems in real time. Phoenix uses a tool called Apache Kafka to move data between its different parts. Kafka works like a shared noticeboard where multiple components, such as the databases, the preprocessing engine, and the AI model, can all read the same information at the same time without interfering with each other.

Kafka was chosen because disasters can cause sudden spikes in data, and Kafka is built to handle that. It can expand to manage large volumes when needed, and if one part of it fails, the rest keeps running because the data is copied across multiple nodes. Another advantage is that the different parts of Phoenix work independently from each other. So if one component slows down, it does not drag the others down with it.

NoFirebase plays a different but complementary role, handling the lighter real-time sync needs of Phoenix's community-facing features things like push notifications, community feedback, and alert delivery, where low latency matters more than raw throughput. Sonnet 4.6

E. Data Preprocessing

Raw data arrives from dozens of sources in dozens of formats and must be transformed into a clean, consistent dataset before the AI models can analyse it effectively. The Data Preprocessing component handles this through three core operations.

Text Normalisation standardises all incoming text lowercasing, removing special characters, expanding abbreviations, and standardising how dates, locations, and organisation names are represented ensuring the same concept is recognised consistently regardless of how it was expressed in the original source.

Duplicate Filtering removes redundant entries before they distort analysis. During disaster events, the same scam message may circulate across thousands of social media accounts. Without deduplication, the AI model would dramatically over-weight a single threat instance. Phoenix uses SimHash-based near-duplicate detection to catch both exact and near-identical duplicates.

NLP Tasks extract structured features from unstructured text through tokenisation, named entity recognition, and part-of-speech tagging. Key entities organisation names, URLs, phone numbers, financial terms, and geographic references are identified and flagged as candidate Indicators of Compromise passed forward to the AI model (Tang et al., 2022).

F. AI Model Processing

The AI Model Processing component is the analytical core of Phoenix. It applies four specialised machine learning tasks to preprocessed data, each targeting a specific dimension of cyber threat behaviour known to emerge during disasters.

Threat Classification uses a transformer-based multi-class classifier fine-tuned against the MITRE ATT&CK taxonomy to categorise content as phishing, ransomware, impersonation scam, fraudulent donation, coordinated misinformation, or benign. Grounding the classifier in MITRE ATT&CK means its outputs are immediately meaningful to cybersecurity professionals and emergency services personnel, rather than requiring additional interpretation (Husari et al., 2017). Scam Detection takes a more targeted approach, applying a dedicated binary classifier trained specifically to recognise the kinds of scams that tend to emerge during disasters think urgent language designed to pressure victims, impersonation of trusted authorities, requests for financial details, and suspicious URLs crafted to look like legitimate government or charity websites.

Scam Detection takes a more focused approach. It uses a separate model that is trained specifically to spot scams that commonly appear during disasters. These scams usually share some clear warning signs, such as urgent language meant to pressure people into acting quickly, someone pretending to be a government body or trusted organisation, requests for money or personal financial details, and

suspicious links designed to look like real government or charity websites. A dedicated scam model (Italian Taxation Office, 2020).

Sentiment Analysis measures the emotional tone of content, with particular attention to fear, panic, and manufactured distress. During disasters, cybercriminals take advantage of the fact that people are scared and not thinking clearly. Phoenix is designed to pick up on these emotional manipulation tactics. This means it can flag suspicious content even when the technical warning signs alone are not strong enough to be certain.

Urgency Detection is specific thing Phoenix looks for is fake urgency. Scammers often use phrases like "Act now" or "Immediate action required" to push people into making quick decisions without thinking. These kinds of phrases show up repeatedly in disaster-related scams.

To catch them, Phoenix combines two methods. The first is a rule-based system that reliably detects known phrases and patterns. The second is a machine learning model that can recognise new variations it has never seen before. Together, they make the detection more accurate and harder to fool. Together, the four tasks produce a multi-dimensional threat profile for each piece of content passed to the Correlation Engine.

G. Correlation Engine

The Correlation Engine is where Phoenix's most consequential analytical work takes place. It takes individual threat profiles from the AI models and applies the most important layer of intelligence the system offers context.

Cross-Referencing Scams and Disasters continuously compares active threat detections against live disaster event data. When a detected threat shares geographic, temporal, or thematic overlap with an active disaster, the engine classifies it as an event-correlated threat. A donation solicitation mentioning a specific affected region appearing at the same moment emergency services have declared a major incident there is treated with an entirely different urgency than the same solicitation appearing months later with no disaster context. This context-sensitivity is what no existing singledomain framework provides (Chakraborty et al., 2024; Salley et al., 2023).

Risk Scoring translates the correlation analysis into a single actionable composite score drawing on four weighted inputs: the severity and scope of the correlated disaster; the strength of thematic overlap between the threat and the event; the AI model's combined classification confidence; and the assessed potential impact on communities or infrastructure. High-scoring threats escalate to DMT human review, medium-scoring threats route through automated triage, and low-scoring content is logged for ongoing monitoring.

H. DMT User Login & Role-Based Access Control

The DMT User Login component provides secure authenticated access for human operators through multi-factor authentication, ensuring only authorised personnel emergency services staff, government analysts, and local government administrators can access the system.

Role-Based Access Control governs what each user can see and do. Read-Only Analysts view risk scores and alerts. DMT Reviewers evaluate and action critical cases. System Administrators configure model parameters and manage data sources. This layered structure ensures no single compromised account can disrupt the alert pipeline (Spyros et al., 2025).

The Community and Auth Feedback module allows verified community members and third-party responders including Police, Fire, Rescue, and Government agencies to submit threat reports and provide feedback on alert accuracy. This realworld ground truth feeds directly into the Model Retraining loop, anchoring Phoenix's continuous improvement to the experiences of the people it protects.

I. DMT Review (Critical Cases)

Every high-risk threat that gets escalated by the Correlation Engine has to go through a human DMT analyst before any public alert is actually sent out. This is a very deliberate choice because if a false positive triggers a public alert, people panics and trust in the system gets damaged. But if a genuine threat gets missed and suppressed, vulnerable people gets left completely exposed. Neither of those outcomes is okay, and having a human reviewer in the loop is exactly what prevents both from happening (Qian et al., 2025; Azambuja et al., 2023).

When reviewing a flagged threat, the DMT analyst looks at the flagged content, the AI model's reasoning, the risk score, and the disaster context before deciding whether to confirm and escalate it to the Alert System, classify it as a false positive for retraining, or defer it for further investigation. Every single decision gets logged and is fully auditable.

For medium-risk cases, the Automation Triage and Dispatcher handles things by routing them directly to the Alert System without needing a full DMT review. This allows Phoenix to respond quickly to high volumes of medium-confidence threats without overwhelming the analysts, while still keeping human accountability firmly in place for the decisions that matters the most.

J. Alert System

So this is basically the part where everything Phoenix has analysed actually gets turned into real action, pushing out verified threat information to communities, emergency services and government authorities as fast and safely as possible.

There is two Alert System instances running in the architecture, one kicks in when a DMT reviewer confirms a threat and the other handles the automated triage channel, but both runs on the same security setup underneath. Alerts goes out through encrypted channels like SMS, email, push notifications via Firebase and platforms like VicEmergency. Everything sent out is authenticated through OAuth 2.0 and locked down with TLS 1.3, meaning nobody can intercept, tamper with or fake it along the way. Sending fake emergency alerts during a disaster is actually a well known attack method, so Phoenix's alert channel has to be something the public can completely trust (Nguyen & Bogle, 2020; Chirgwin, 2023).

Every alert gets saved into an audit log that cannot be changed or deleted, so there is always a solid record of what was sent and when. And if a delivery fails just because of network is struggling, the system keeps it on automatically retrying until the alert actually gets through to whoever needs it.

K. Feedback Collection & Model Retraining

Cyber adversaries do not stay still they adapt quickly as initial attack patterns are detected and blocked. A Phoenix system that does not continuously learn would inevitably fall behind. The Model Retraining loop is the architectural response to this reality.

Feedback Collection gathers improvement signals from DMT reviewer decisions identifying confirmed threats and false positives, community and third-party reports submitted through the Auth Feedback module, and automated system logs tracking alert delivery and response outcomes.

Model Retraining uses this feedback to update the AI models through online learning or scheduled batch retraining. Novel scam types and attack patterns encountered during a disaster are incorporated into the training dataset so Phoenix recognises them immediately if they reappear. All retraining is version-controlled with rollback capability.

Discard if Irrelevant filters out low-quality, ambiguous, or adversarially crafted feedback before it enters the training pipeline. Feedback manipulation deliberately submitting misleading reports to degrade model performance is itself a viable attack vector, and Phoenix treats its training data with the same scepticism it applies to incoming threat data (Azambuja et al., 2023; Spyros et al., 2025).

DMT Received by Community or Third Party represents the highest-confidence feedback signal in the system. Reports from Police, Fire, Rescue, and Government agencies carry authoritative verification that threats are real. These groundtruth signals are weighted accordingly in the retraining pipeline, accelerating Phoenix's ability to recognise and respond to the threat patterns that matter most.

L. Store & Retrieve Operations

Throughout the Phoenix flowchart, **Store** and **Retrieve** operations appear at specific points connecting the pipeline components to the Scam DB and Cyber Threat DB. These are not simply technical labels they represent deliberate decisions about when data should be saved and when it should be recalled.

Stream Processing → Store → Scam DB (Left Side) writes a copy of validated incoming cyber threat and scam data into the Scam DB immediately after stream processing. Storing at this early stage ensures no raw threat data is ever lost and continuously grows the historical scam repository that the AI model draws upon later.

Stream Processing → Store → Scam DB / Cyber Threat DB (Right Side) simultaneously writes broader cyber threat intelligence — indicators of compromise, known attack patterns, and MITRE ATT&CK-mapped adversary tactics — into both databases. This early storage ensures the data is available to downstream components the moment they need it.

AI Model Processing → Store → Scam DB writes the AI model's enriched, classified outputs — threat labels, scam detection results, sentiment scores, and urgency flags — back into the Scam DB. This creates both a permanent auditable record of every model decision and a growing labelled dataset for future model retraining.

Scam DB / Cyber Threat DB → Retrieve → AI Model Processing pulls relevant reference data from both databases before the AI models classify incoming content. This is what allows the models to compare new content against historical records of confirmed scams and known attack patterns — recognising a new scam that follows the same structure as a previously documented one even if the specific wording has changed (Li et al., 2022).

Scam DB / Cyber Threat DB → Retrieve → Correlation Engine supplies the Correlation Engine with historical threat records when performing its cross-referencing and risk scoring. This historical context allows the engine to score threats based not just on current characteristics but on how

similar threats have behaved and escalated during previous disaster events (Li et al., 2022; Huang & Xiao, 2024).

Together, Store and Retrieve operations form a continuous intelligence loop. Every Store enriches the databases that future Retrieves draw from. Every Retrieve makes the AI model and Correlation Engine more accurate. Over successive disaster events, this relationship creates a compounding effect — Phoenix becomes progressively smarter with every event it processes.

Identify All Entry Points

When building a system like Phoenix, one of the most important questions a cybersecurity team must ask is not just how the system works but where it is open to the world. Every location where data, commands, or users can enter from an external or untrusted source is called an entry point, and every entry point is, by definition, a place where something could go wrong. Understanding these points is not optional — it is the starting place for every security decision made in the architecture (Chakraborty et al., 2024).

Phoenix is a system that deliberately reaches outward. It pulls in disaster feeds, monitors social media, accepts community reports, and pushes out public alerts. That connectivity is what gives it power. But it also means the team has a responsibility to look honestly at every door that has been opened and ask what happens if someone tries to come through that door uninvited?

Six entry points have been identified in the Phoenix architecture. Each one is examined below with a clear explanation of what enters the system at that point, where it comes from, why it matters from a security perspective, and what could realistically go wrong if it is not properly protected.

Entry Point 1 — External Disaster Data APIs

What enters: Real-time disaster event data, weather alerts, flood warnings, bushfire notifications, and seismic reports.

Source: Bureau of Meteorology (BoM), Country Fire Authority (CFA), Emergency Management Victoria, and Geoscience Australia via REST APIs.

Why it matters: Phoenix's entire contextual threat detection depends on this feed being accurate. When a disaster is detected here, the whole system elevates its monitoring sensitivity for that region. An adversary who understands this does not need to break into Phoenix, they only need to manipulate what Phoenix sees. A corrupted feed could trigger false alarms or, more dangerously, prevent threat elevation during a real event when communities need protection most (Chakraborty et al., 2024).

Attack risks: API key compromise; man-in-the-middle attacks injecting false disaster data; denial-of-service attacks cutting off the intelligence pipeline; DNS spoofing redirecting API calls to malicious servers.

Entry Point 2 — External Cyber Threat & Scam Intelligence Feeds

What Enters: So basically what comes into this entry point is a whole mix of things. You have got scam reports, open-source intelligence, MITRE ATT&CK threat indicators, social media content, and suspicious activity that has been submitted by members of the community.

Source: All of this is coming in from places like ACCC Scamwatch, the Australian Signals Directorate, the MITRE ATT&CK database, Twitter/X API, Facebook API, and various other open-source intelligence feeds that Phoenix monitors.

Why It Matters: This is honestly the widest and most open entry point in the whole Phoenix system. It is taking in everything from really authoritative government intelligence all the way down to completely unverified posts that random people has put up on social media. And that is exactly what makes it so risky. An attacker does not even need to go anywhere near Phoenix directly to cause damage here. All they needs to do is start introducing enough manipulated or false content into the public sources that Phoenix is already monitoring, and over time that starts to quietly influence what the AI model is seeing and learning from. It is a slow and subtle kind of attack that is really hard to spot and by the time anyone notices something is wrong, the damage has already been building up for a while (Ghazi et al., 2018; Li et al., 2022).

Attack Risks: The main risks here includes things like data poisoning through fake scam report submissions, adversarial content that has been carefully crafted to slip past the AI detection without getting flagged, API rate-limiting attacks that disrupts the flow of data coming into the system, and deliberate flooding with massive amounts of fake high-priority indicators that is designed to bury the genuine threats and completely overwhelm the DMT analysts who is trying to keep up with everything.

Entry Point 3 — DMT User Login Portal

What enters: User credentials, authentication tokens, and session data from human operators.

Source: Disaster Management Team personnel emergency services staff, government analysts, and local government administrators.

Why it matters: This is the entry point with the most direct human consequence. People who log in here approve public alerts, review critical cases, and configure the AI models. A compromised account does not just expose data, it puts Phoenix's controls in the wrong hands. It is also worth acknowledging that DMT staff working under the pressure of an active disaster are more vulnerable to phishing and social engineering than under normal conditions. Attackers know this and deliberately plan around it (Azambuja et al., 2023).

Attack risks: Brute force and credential stuffing attacks; targeted phishing campaigns against DMT staff during active disasters; session hijacking through stolen authentication tokens; insider threat misuse of legitimate accounts.

Entry Point 4 — Community and Auth Feedback Module

What enters: Threat reports, alert feedback, and verification data from community members and verified responders.

Source: General public, Police, Fire, Rescue services, and Government agencies.

Why it matters: This is the most open-facing entry point in Phoenix, deliberately accessible to a wide range of users including the general public. That openness is necessary the feedback collected here drives model retraining. But it also creates a real tension. The most dangerous attack here is not loud or obvious — it is quiet and cumulative, gradually shifting what the model understands a threat to look like until Phoenix begins missing the very things it was built to catch (Hunt et al., 2022; Vicari & Komendatova, 2023).

Attack risks: Spam flooding overwhelming the feedback pipeline; adversarial feedback corrupting the retraining dataset over time; impersonation of verified responders injecting false high-confidence signals; coordinated disinformation campaigns systematically shifting model behaviour.

Entry Point 5 — Third-Party DMT Reports (Police, Fire, Rescue, Government)

What enters: Authoritative threat reports, incident confirmations, and verified intelligence from official emergency services and government agencies.

Source: Victoria Police, Country Fire Authority, State Emergency Service, relevant government departments, and authorised third-party responders.

Why it matters: Reports from official agencies carry the highest confidence weighting in the Phoenix retraining pipeline, they are treated as verified ground truth. That elevated trust is necessary, but it also means a successfully impersonated official submission would be acted upon quickly and weighted heavily in model retraining. The damage from a single well-crafted false submission from a spoofed official source could ripple through the system's behaviour long before the compromise is detected (Salley et al., 2023; Chakraborty et al., 2024).

Attack risks: Impersonation of official agencies through spoofed credentials; injection of false incident confirmations manipulating the Correlation Engine; interception and silent alteration of legitimate submissions in transit.

Entry Point 6 — Alert System API Endpoints

What enters: Outbound alert delivery commands, authentication tokens, and message content sent to external alert delivery channels.

Source: Internal Phoenix Alert System components triggering deliveries to SMS gateways, email servers, Firebase, and platforms such as VicEmergency.

Why it matters: This entry point is different from all others, it is about what goes out rather than what comes in. What goes out here are official emergency alerts delivered to people who are already frightened and relying on those messages to make safety decisions. An attack here does not target what Phoenix knows — it targets what Phoenix does. The consequences are not measured in data or system performance. They are measured in public trust and the real harm that comes from either a suppressed warning or a false one reaching thousands of people during a disaster (Nguyen & Bogle, 2020; Chirgwin, 2023).

Attack risks: Interception and alteration of alert API calls in transit; replay attacks re-executing old commands at wrong times; alert suppression silently blocking critical warnings from reaching communities; spoofing of the Phoenix alert channel to issue fake official alerts carrying false or harmful instructions.

Summary of Entry Points

Entry Point	Source	Primary Risk
External Disaster Data APIs	BoM, CFA, Emergency Management Victoria	Feed manipulation, API spoofing
Cyber Threat & Scam Intelligence Feeds	Scamwatch, ASD, Social Media APIs	Data poisoning, adversarial content injection
DMT User Login Portal	DMT staff and administrators	Credential theft, phishing, session hijacking
Community & Auth Feedback Module	Public and verified responders	Spam flooding, adversarial feedback, impersonation
Third-Party DMT Reports	Police, Fire, Rescue, Government	Agency impersonation, report injection
Alert System API Endpoints	Internal Phoenix alert components	Interception, replay attacks, alert suppression

Why Entry Point Analysis Matters for Phoenix

Every protective measure in Phoenix traces back to a specific entry point risk. Data Validation exists because of Entry Points 1 and 2. Multi-factor authentication exists because of Entry Point 3. The Discard if Irrelevant filter exists because of Entry Points 4 and 5. TLS 1.3 encryption and OAuth 2.0 authentication exist because of Entry Point 6. Phoenix reaches out into the world to protect it — and that reach is both its greatest strength and its greatest exposure. Identifying and understanding every entry point is what ensures that openness never becomes a vulnerability (Chakraborty et al., 2024; Azambuja et al., 2023).

Identifying Trust Boundaries

If entry points tell us where data and users enter Phoenix, trust boundaries tell us where the system stops and asks — should I actually trust this? A trust boundary is a deliberate line drawn within the architecture where data or control passes from one zone to another, and where Phoenix must make a conscious decision about whether what is crossing that line has earned the right to proceed.

In a system like Phoenix, trust is not given freely. A validated government API feed is not the same as an anonymous community report. An authenticated DMT administrator action is not the same as a public submission. Recognising where these differences matter and enforcing the right controls at the right boundaries is what keeps Phoenix reliable even when parts of its environment are working against it (Azambuja et al., 2023).

Five trust boundaries have been identified in the Phoenix architecture. Each is explained below with a clear account of where it sits, what crosses it, why the trust difference matters, and what enforces it.

Trust Boundary 1 — External World → Data Validation

Where it sits: Between all external data sources and the Phoenix internal pipeline.

What crosses it: Disaster event feeds, cyber threat intelligence, scam reports, and social media data arriving from external APIs and open-source sources.

The trust difference: Everything outside this boundary is untrusted by default. No external source however authoritative it appears that can be assumed to be accurate, authentic, or uncompromised at any given moment. The moment data is allowed past this boundary, the system begins treating it as something it can work with. That shift from untrusted to trusted must be earned through verification, not assumed through familiarity (Chakraborty et al., 2024).

What enforces it: The Data Validation component stands as the gatekeeper here. Schema validation, format checking, source authentication through API keys and digital certificates, and deduplication are all applied before any data proceeds. If data cannot be fetched at all, a Failure to Fetch flag is raised and the retry mechanism activates rather than allowing the pipeline to move forward on incomplete information. Nothing crosses this boundary without being checked.

Why it matters: This is the widest and most consequential trust boundary in Phoenix. Every piece of data the system ever acts on enters here first. A failure at this point a spoofed feed accepted as legitimate, a poisoned dataset passed through unchecked would send corrupted information through every downstream component. This boundary has to be right because everything else depends on it (Salley et al., 2023).

Trust Boundary 2 — Stream Processing → Internal Databases (Scam DB / Cyber Threat DB)

Where it sits: Between the Stream Processing layer and the Scam DB and Cyber Threat DB.

What crosses it: Validated, structured data being written into the internal databases for storage and future retrieval.

The trust difference: Data that has cleared the first boundary has earned a degree of trust it has been authenticated and structurally verified. But writing data into a database introduces a new risk. A database that holds corrupted or manipulated records does not just affect one processing cycle it affects every future AI model classification and every Correlation Engine risk assessment that draws on those records (Li et al., 2022). The databases are long-term assets, and what goes into them shapes how Phoenix behaves for as long as it operates.

What enforces it: Input sanitisation before database writes protects against injection attacks that could corrupt stored records or expose database contents. Write access is restricted to authorised internal pipeline components only no external source can write directly to either database without passing through the validated pipeline first. Data integrity checks confirm that records being written match expected formats before they are committed.

Why it matters: The databases are Phoenix's memory. They hold the historical scam records, threat indicators, and classified AI outputs that the system draws on every time it analyses new content. Corrupted memory produces corrupted judgements and in a system making decisions that affect public safety during disasters, corrupted judgements carry real consequences (Li et al., 2022; Huang & Xiao, 2024).

Trust Boundary 3 — AI Model Processing → Correlation Engine

Where it sits: Between the AI Model Processing component and the Correlation Engine.

What crosses it: Multi-dimensional threat profiles threat classification labels, scam detection results, sentiment scores, and urgency flags generated by the AI models for each piece of processed content.

The trust difference: The AI models are powerful, but their outputs are probabilistic. Every classification comes with a confidence score reflecting how certain the model is. Passing a low-confidence output to the Correlation Engine without flagging its uncertainty would be the same as treating a guess as a fact (Qian et al., 2025). The Correlation Engine needs to know not just what the AI model concluded, but how confident it was because that confidence is a critical input to the risk scoring process that determines how every threat is handled.

What enforces it: Confidence threshold validation ensures that outputs falling below a defined certainty level are flagged before being passed forward, triggering additional scrutiny rather than being accepted at face value. Adversarial input detection monitors for outputs that deviate significantly from expected patterns a potential sign that the model has been fed crafted content designed to manipulate its classifications. Output validation confirms that threat profiles conform to expected structures before the Correlation Engine acts on them.

Why it matters: The Correlation Engine is where threat profiles become risk scores and risk scores become routing decisions. A manipulated or low-confidence AI output accepted without question at this boundary could cause a genuine threat to be under-scored and missed, or a false positive to be over-escalated unnecessarily. Both outcomes undermine the reliability Phoenix exists to provide (Azambuja et al., 2023; Qian et al., 2025).

Trust Boundary 4 — DMT Login → Internal System Functions

Where it sits: Between the public-facing DMT login portal and the privileged internal functions of the Phoenix platform.

What crosses it: Authenticated user sessions granting access to threat review, alert approval, model configuration, and system administration functions.

The trust difference: The DMT login portal is accessible from outside the system it has to be. The functions behind it are not. Reviewing critical threat cases, approving public alerts, configuring AI model parameters, and managing data sources are all actions with direct real-world consequences. This boundary separates what anyone can attempt from what only verified and authorised personnel can do (Spyros et al., 2025) and it is one of the most important lines in the entire Phoenix architecture.

What enforces it: Multi-factor authentication is the primary control, ensuring a stolen password alone is not enough to cross this boundary. Role-Based Access Control limits what each authenticated user can see and do once inside — containing the damage if any single account is compromised. Session management controls including timeout policies and token expiry prevent authenticated sessions from being left open indefinitely or reused after they end. Anomalous login detection monitors for unusual access patterns and raises alerts when something does not look right.

Why it matters: A breach at this boundary does not just compromise data it compromises agency. An adversary who crosses this line gains the ability to act as Phoenix making decisions that the system and the public will treat as legitimate. During an active disaster, that is an extraordinarily dangerous position to be in (Azambuja et al., 2023; Spyros et al., 2025). Protecting this boundary with multiple overlapping controls is not overcaution. It is proportionate to what is at stake.

Trust Boundary 5 — Alert System → Public Communication Channels

Where it sits: Between the internal Phoenix Alert System and external public-facing communication channels SMS gateways, email servers, Firebase push services, and platforms such as VicEmergency.

What crosses it: Verified alert messages carrying official threat intelligence and emergency notifications to communities, emergency services, and government authorities.

The trust difference: Inside Phoenix, an alert is a data object a structured message that has been reviewed, approved, and queued for delivery. The moment it crosses this boundary and enters a public communication channel, it becomes something entirely different. It becomes an official statement received by people who are frightened, who are making real decisions based on what they read, and who have no practical way of knowing whether it is genuine. The trust that communities place in official emergency alerts is not a technical property it is a social contract (Nguyen & Bogle, 2020). And this boundary is where Phoenix either honours that contract or breaks it.

What enforces it: All outbound communications are authenticated using OAuth 2.0 to verify that delivery commands originate from legitimate Phoenix components, and encrypted using TLS 1.3 to prevent interception or alteration in transit. An immutable audit log records every alert action what was sent, when, to whom, and by which component creating a permanent accountability trail. Automatic retry logic with exponential backoff ensures delivery failures during network degradation do not silently prevent alerts from reaching recipients. Rate limiting prevents replay attacks from triggering mass duplicate alert delivery.

Why it matters: Public trust in official emergency alerts takes years to build and moments to destroy. A single successful attack at this boundary a suppressed warning, a false alert, or a spoofed message does not just cause immediate harm. It plants doubt in the minds of communities about whether official alerts can be trusted at all, making every future warning less effective and every future disaster more dangerous (Nguyen & Bogle, 2020; Chirgwin, 2023). Protecting this boundary is not just a technical obligation. It is a civic one.

Summary of Trust Boundaries

Trust Boundary	Sits Between	Key Controls
External World → Data Validation	External sources and internal pipeline	Schema validation, source authentication, deduplication
Stream Processing → Databases	Validated data stream and internal databases	Input sanitisation, restricted write access, integrity checks
AI Model → Correlation Engine	AI model outputs and risk scoring	Confidence threshold validation, adversarial input detection
DMT Login → Internal Functions	Public login portal and privileged system functions	MFA, RBAC, session management, anomalous login detection
Alert System → Public Channels	Internal alert components and public communication	OAuth 2.0, TLS 1.3, audit logging, retry logic, rate limiting

Why Trust Boundaries Matter for Phoenix

Entry points tell Phoenix where to watch. Trust boundaries tell it what to believe. Together they define the security posture of the entire system not as a collection of isolated controls, but as a coherent layered defence where every part of the architecture knows exactly how much trust to extend to what arrives from outside it.

Each boundary identified here sits at a point where misplaced trust carries significant consequences where accepting something that should have been questioned could corrupt a database, manipulate a risk score, grant unauthorised access, or undermine the integrity of a public alert. Phoenix has been built with the understanding that trust in a security context is not a feeling. It is a decision one that must be earned at every boundary, enforced consistently, and never taken for granted (Azambuja et al., 2023; Chakraborty et al., 2024).

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